

AI & ML in speech systems

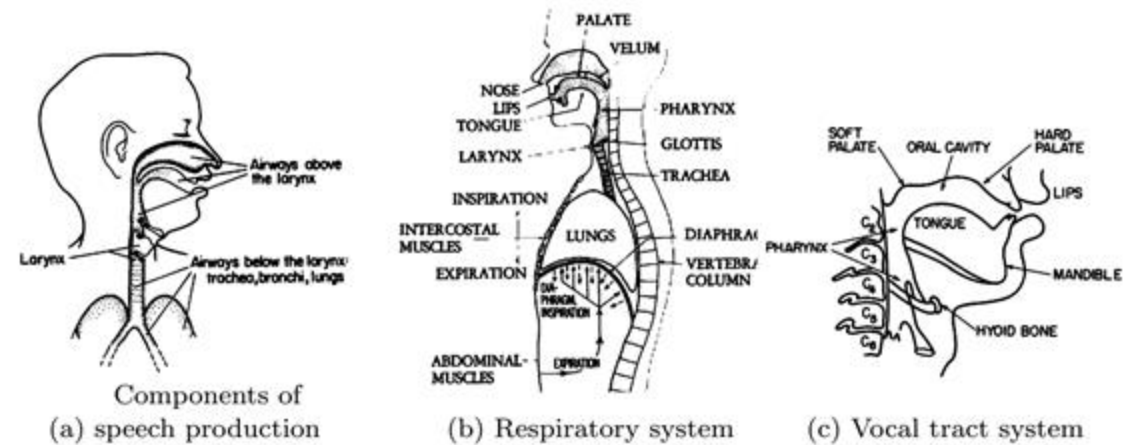
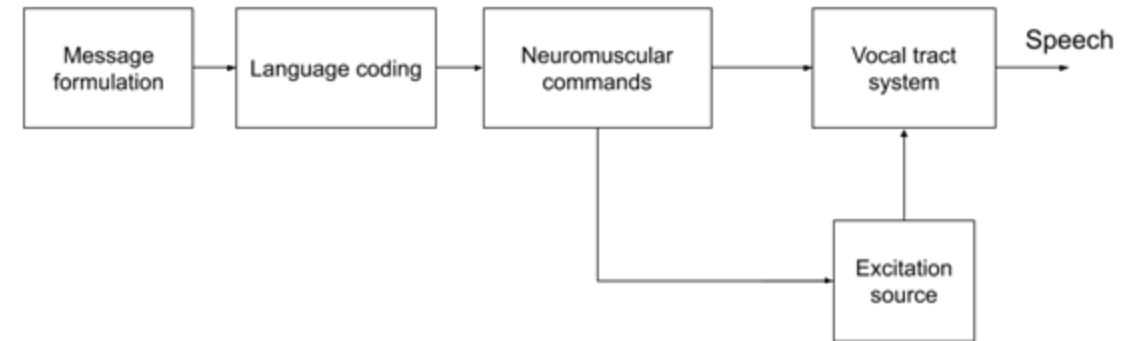
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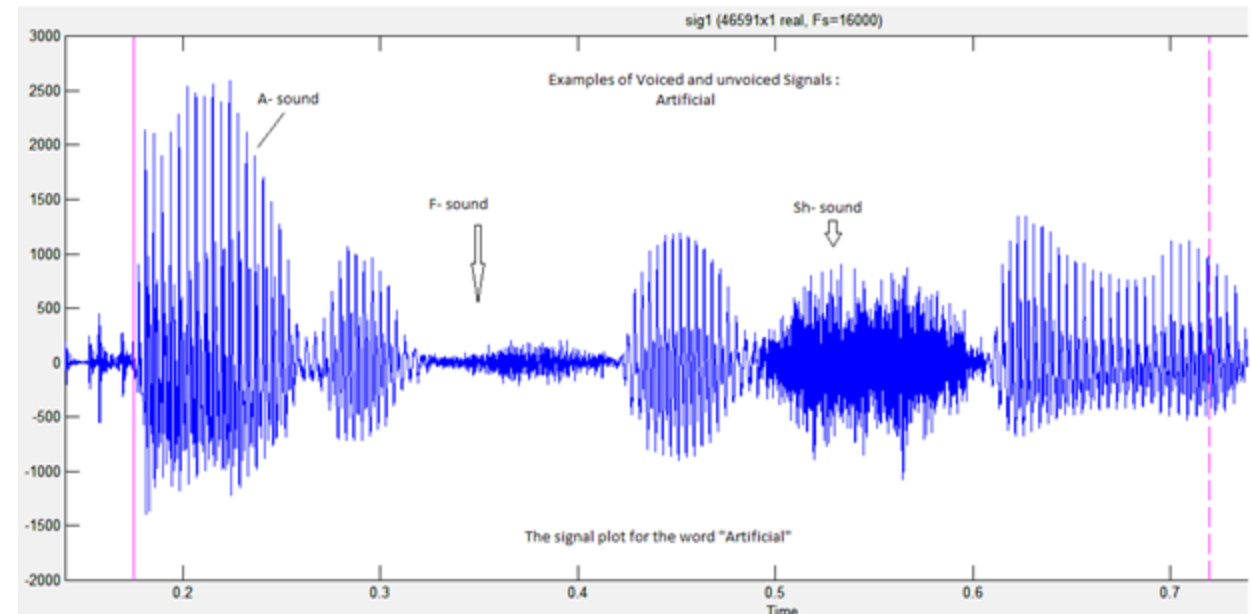
Introduction

- **Speech** is a unique, complex, and dynamic motor activity through which we express our thoughts and emotions.
- Natural mode of communication for human beings.
- Important in human computer interaction



Introduction contd.

- Information in speech
 - Message
 - Language
 - Gender
 - Age
 - Speaker identity
 - Emotional state
 - Cognitive behaviors of
 - Depression
 - Autism
 - Language dis
 - Abnormalities in speech production etc.



Typical speech signal for a word "Artificial"

Introduction contd.

- **Applications of speech processing:**

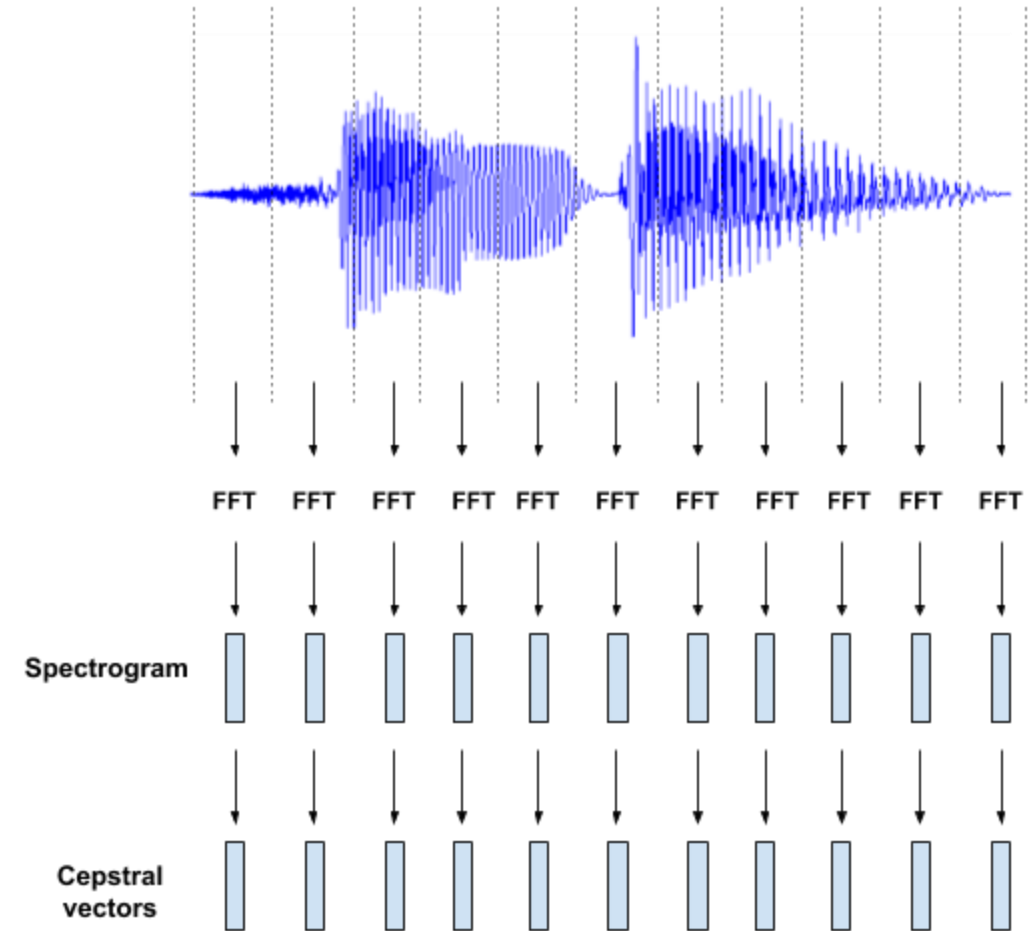
- Automatic speech recognition
- Text-to-speech synthesis
- Spoken language dialog systems
- Speaker recognition/verification
- Emotion recognition
- Language identification
- Pronunciation evaluation for clinical tools
- Forensic tools
- Speech coding

- **Importance of speech technologies:**

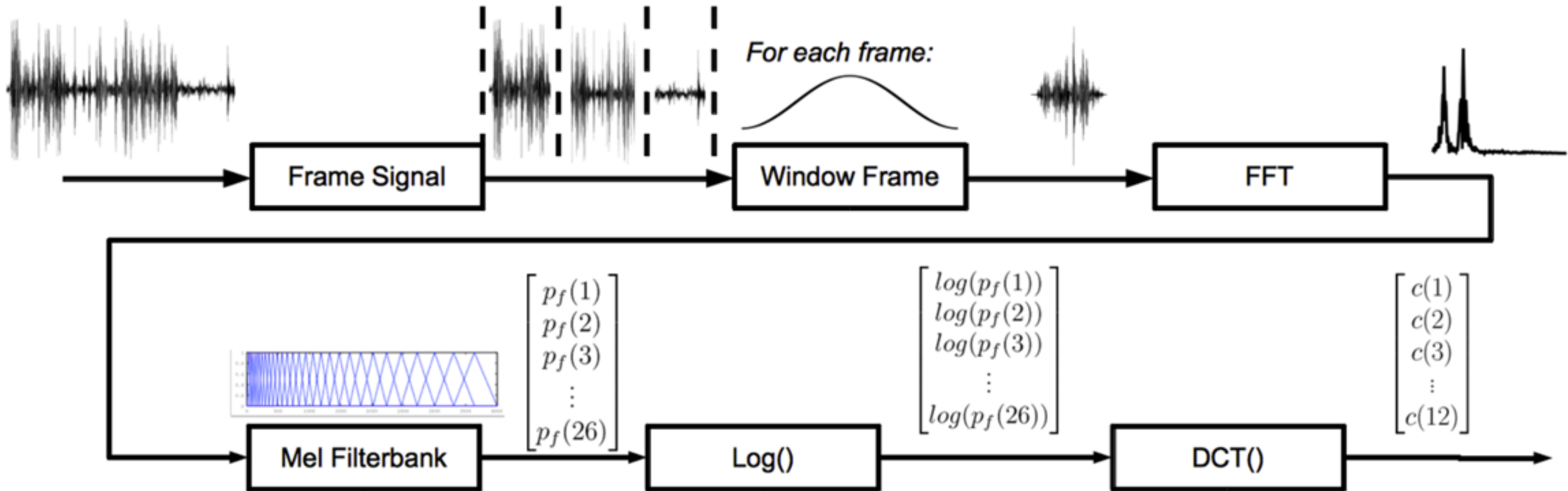
- For better human machine interaction
- For information security and authentication
- Speech based smart appliances
- Multilingual speech systems (speech to speech translation)
- As a navigator for disabled people
- As a personalized speech therapist
- As a personalized assistant in education, agriculture, health care, and other service sectors.

Feature extraction

- Primary step in the development of speech systems
- Fourier transform
- Short time Fourier analysis due to non-stationarity
- 20-30 ms frame size of speech
- Window type: Hamming, Hanning, Rectangular etc.
- Cepstral coefficients are most common feature representation the which captures the vocal tract shape and dynamic information
- Mel-frequency cepstral coefficients (MFCC), Linear prediction cepstral coefficients (LPCC), Perceptual linear prediction cepstral coefficients (PLPCC), etc.



Feature extraction contd.



- Including the energy, total 13-dimensional static cepstral features
- 39-dimensional MFCC features: [static -- delta -- delta-delta]

Speech systems

- **Speaker recognition:** is a technique to recognize the identity of a speaker from a speech utterance.
- **Speaker verification** aims to verify whether an input speech corresponds to the claimed identity.
- **Emotion recognition** is a process of identifying the emotional state (Anger, Happy, Sad, Neutral, Fear, etc.) of a speaker from the spoken utterance.
- **Spoken Language identification** is a task of recognizing the language identity/information from the speech signal.
- **Automatic speech recognition** is the process of deriving the transcription (word sequence) of an utterance, given the speech waveform.
- **Text to speech synthesis** aims to convert the given text (word sequence) information into speech.

Speech systems contd.

Speaker identification/verification

Language identification

Emotion recognition

Simple classification problems

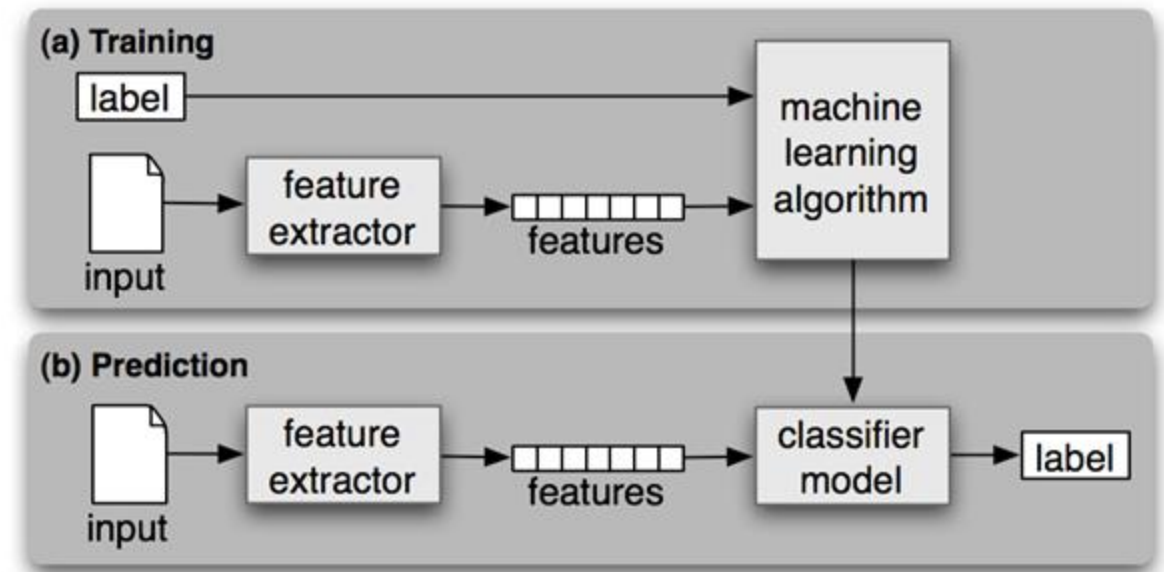
Automatic speech recognition (ASR)

Text to speech synthesis

Are advanced speech systems require:
classification, duration/pronunciation
modelling, sequence-sequence mapping etc.

Speech systems contd.

- **Input:** speech signal
- **Features:** 39-dimensional MFCCs
- **Class labels:**
 - Language ID (Ex: Telugu-1, Hindi-2, English-3, Tamil-4, etc.) in language identification task. Similarly, speaker ID will be the class label in speaker identification.
- **Machine learning algorithms:**
 - Gaussian Mixture Modelling (GMM), GMM with universal background modelling, I-vector modelling.



Classification system example

Gaussian Mixture Modelling (GMM)

Parameter estimation:

Assume we have a collection of acoustic frames $X = \{\mathbf{x}_t\}_{t=1}^T$ for estimation of model parameters θ

Maximum likelihood (ML) estimation

$$\theta_{\text{ML}} = \arg \max_{\theta} p(X|\theta)$$

Maximum *a posteriori* (MAP) estimation

$$\theta_{\text{MAP}} = \arg \max_{\theta} p(\theta|X) = \arg \max_{\theta} p(X|\theta)p(\theta)$$

where $p(\theta)$ denotes the prior distribution of θ

Gaussian Mixture Modelling (GMM) contd.

Formulation of GMM:

Gaussian mixture model (GMM) is a **weighted sum** of Gaussians

$$p(\mathbf{x}|\boldsymbol{\theta}) = \sum_{i=1}^M \pi_i b_i(\mathbf{x})$$

$$\boldsymbol{\theta} = \{\pi_i, \mathbf{u}_i, \Sigma_i\}$$

π_i : mixture weight

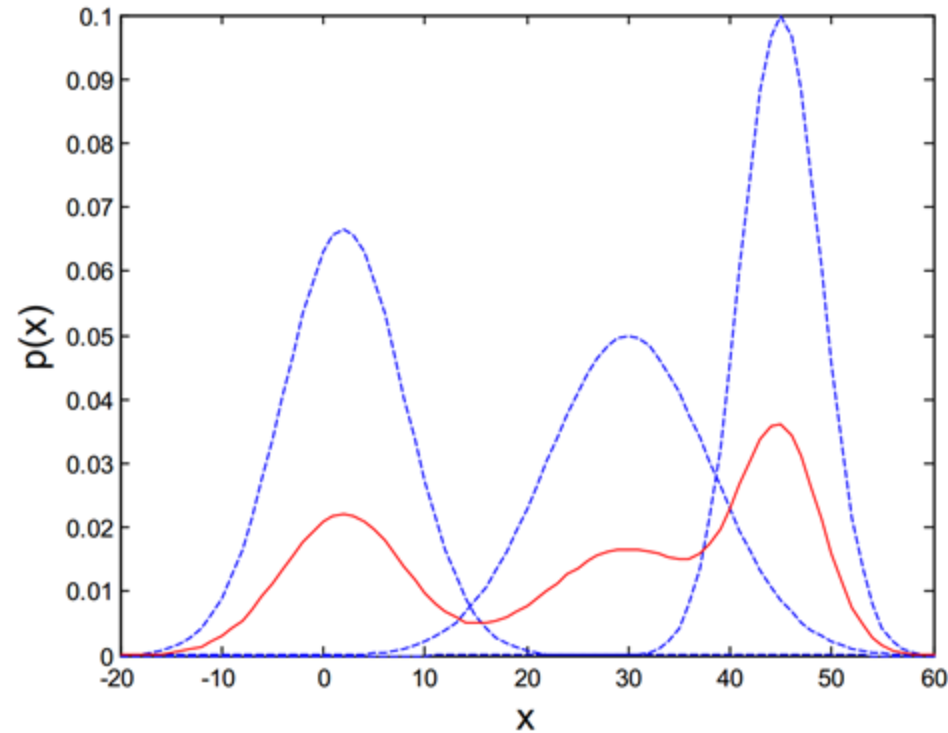
$\mathbf{u}_i$: mixture mean vector

Σ_i : mixture covariance matrix

$$b_i(\mathbf{x}) = \frac{1}{(2\pi)^{D/2} |\Sigma_i|^{1/2}} \exp \left(-\frac{1}{2} (\mathbf{x} - \mathbf{u}_i)^\top \Sigma_i^{-1} (\mathbf{x} - \mathbf{u}_i) \right)$$

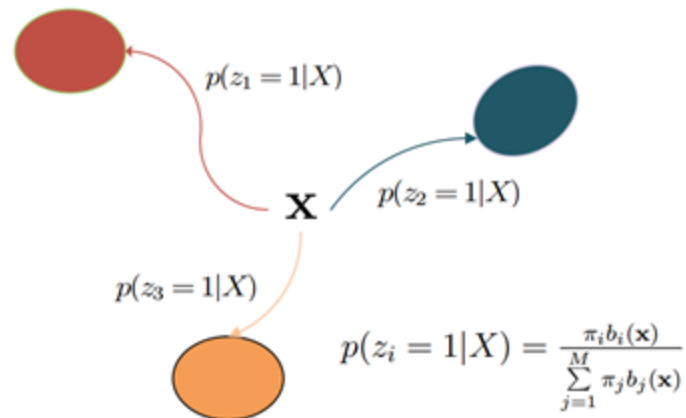
Mixture component z_i is a **latent variable** which is either zero or one

Gaussian Mixture Modelling (GMM) contd.



GMM distribution from three Gaussians

Gaussian Mixture Modelling (GMM) contd.



Accumulate **sufficient statistics**

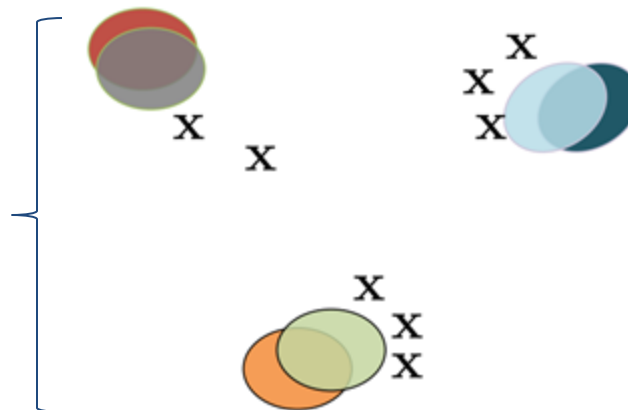
$$T_i = \sum_{t=1}^T p(z_{ti} = 1|X)$$

$$\mathbb{E}_i(\mathbf{x}) = \sum_{t=1}^T p(z_{ti} = 1|X) \mathbf{x}_t$$

$$\mathbb{E}_i(\mathbf{x}\mathbf{x}^T) = \sum_{t=1}^T p(z_{ti} = 1|X) \mathbf{x}_t \mathbf{x}_t^T$$

E-step

M-step



GMM parameters

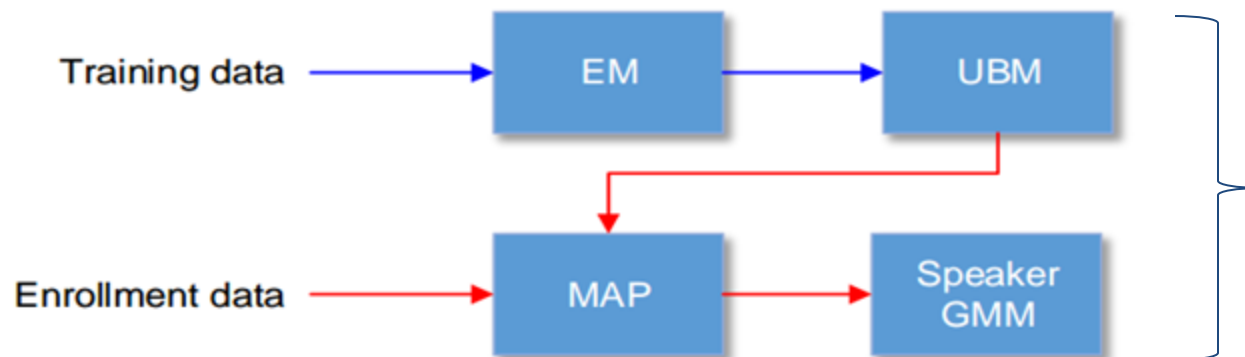
$$\pi_i^{\text{new}} = \frac{T_i}{T}$$

$$\mu_i^{\text{new}} = \frac{1}{T_i} \mathbb{E}_i[\mathbf{x}]$$

$$\Sigma_i^{\text{new}} = \frac{1}{T_i} \mathbb{E}_i[\mathbf{x}\mathbf{x}^T] - \mu_i \mu_i^T$$

GMM with universal background model (GMM-UBM)

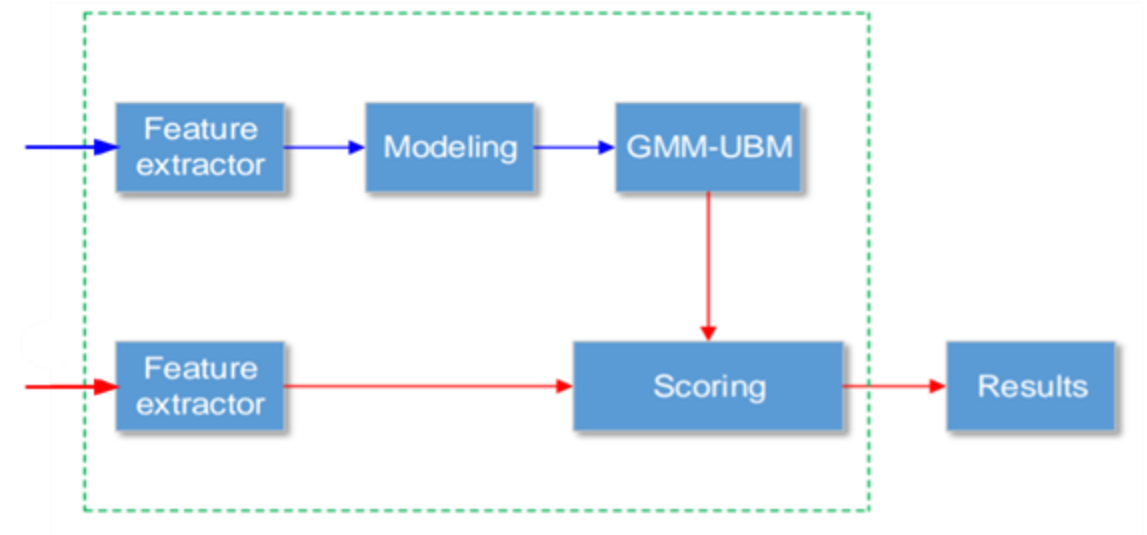
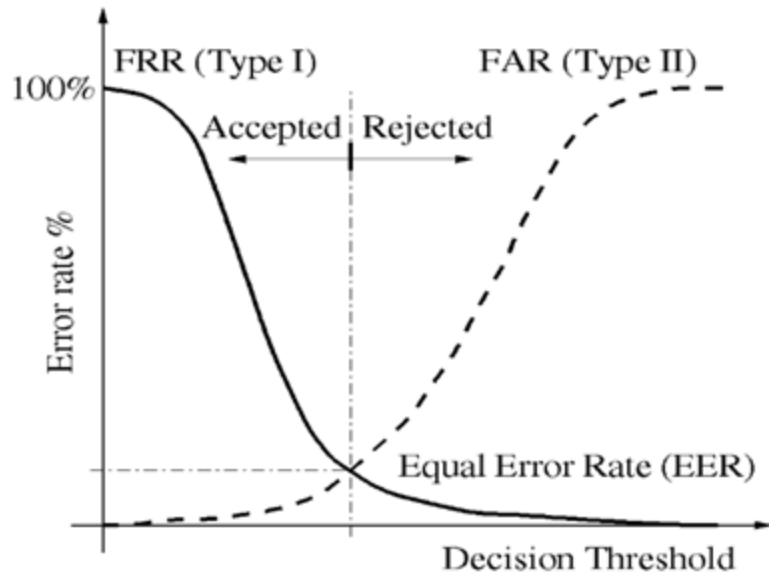
- GMM are used for both **target** and **background models**
 - **Target model** of a class is trained using features corresponding to that class
 - **Universal background model** is trained using features from many classes
- Target model is adapted from universal background model (UBM)
 - good with limited target training data
- **Maximum a posteriori (MAP) adaptation:**
 - align target training vectors to UBM
 - accumulate sufficient statistics
 - update target model parameters with smoothing to UBM parameters



GMM-UBM Example:
Adjustment of enrolled speaker's GMM using UBM.

GMM-UBM based classification

Metrics: Accuracy, and Equal error rate (EER). EER is function of false acceptance ratio (FAR) and False rejection ratio (FRR).



Procedure for speaker recognition, language identification, emotion recognition using GMM-UBM modelling

Joint factor analysis and I-vector modelling

- Factor analysis is a statistical method which is used to describe the variability among the observed variables in terms of potentially lower number of unobserved variables called factors.
- Joint factor analysis (JFA) was the initial paradigm for **speaker recognition**
- Later, it is used in Language identification, Emotion recognition, Automatic speech recognition, etc.
- **Intuition and interpretation**
 - A supervector for a speaker (language / emotion) should be decomposable into speaker independent, speaker dependent, channel dependent, and residual components
 - Each component is represented by low-dimensional factors, which operate along the principal dimensions of the corresponding component
 - Speaker (language/emotion) dependent component, known as the eigen voice, and the corresponding factors

Joint factor analysis and I-vector modelling

GMM supervector $\mathbf{u}$ for a speaker (language/emotion/ any other) can be decomposed as:

$$\mathbf{u} = \mathbf{m} + \mathbf{V}\mathbf{y} + \mathbf{U}\mathbf{x} + \mathbf{D}\mathbf{z}$$

The diagram illustrates the decomposition of the GMM supervector $\mathbf{u}$ into four components. The equation $\mathbf{u} = \mathbf{m} + \mathbf{V}\mathbf{y} + \mathbf{U}\mathbf{x} + \mathbf{D}\mathbf{z}$ is shown. Arrows point from labels to the corresponding terms: 'Speaker supervector' points to $\mathbf{u}$, 'Speaker-independent component' points to $\mathbf{m}$, 'Speaker-dependent component' points to $\mathbf{V}\mathbf{y}$, 'Channel-dependent component' points to $\mathbf{U}\mathbf{x}$, and 'Speaker-dependent residual component' points to $\mathbf{D}\mathbf{z}$.

where

- $\mathbf{m}$ is a **speaker-independent** supervector from UBM
- $\mathbf{V}$ is the **eigenvoice** matrix
- $\mathbf{y} \sim \mathcal{N}(0, \mathbf{I})$ is the **speaker factor** vector
- $\mathbf{U}$ is the **eigenchannel** matrix
- $\mathbf{x} \sim \mathcal{N}(0, \mathbf{I})$ is the **channel factor** vector
- $\mathbf{D}$ is the **residual** matrix, and is diagonal
- $\mathbf{z} \sim \mathcal{N}(0, \mathbf{I})$ is the **speaker-specific** residual factor vector

Joint factor analysis and I-vector modelling

- I-vectors gives the utterance level representation. Cosine distance can be used to find similarity between I vectors.
- Variable length sequence to fixed dimension representation
- I-vectors with **SVM** or **DNN** or **PLDA** scoring are used for speaker/emotion/language identification

I-vector modelling in Language identification

- **Motivation**

- I-vector models are the state-of-art baseline models in NIST 2009 Language Recognition Evaluation (**NIST 2009 LRE**) and Oriental language recognition (**OLR**) challenge.
- Allows low dimensional speech representation based on the Factor analysis
- Each speech recording is mapped on low dimensional vector (Ex: 400)
- Factor analysis as feature extractor
- Modeling the inter-language variability between different language classes

I-vector modelling in Language identification

- C_{avg} is the average pairwise loss between miss rate and false alarm rate;
- **EER** refers to equal error rate.
- Low values of these metrics implies better system performance

C_{avg} AND EER RESULTS OF VARIOUS I-VECTOR BASELINE SYSTEMS ON 3 TEST CONDITIONS.

	1 second		3 second		Full-Length	
System	C_{avg}	EER%	C_{avg}	EER%	C_{avg}	EER%
i-vector	0.1672	15.28	0.0695	7.59	0.0522	6.224
i-vector + LDA	0.1238	13.30	0.0494	5.95	0.0362	4.704
i-vector (Linear SVM)	0.1485	14.43	0.0624	6.07	0.0469	4.58
i-vector (Poly SVM)	0.1242	12.43	0.0470	4.83	0.0351	3.58
i-vector (RBF SVM)	0.1313	12.16	0.0495	4.59	0.0352	3.39

I-vector modelling in Emotion recognition

- **INTERSPEECH 2009 Emotion recognition Challenge:**
 - uses **FAU-AIBO** Emotion corpus, It is two class problem: **Positive** or **Negative**
- **GMM-MFCC systems (s1)**
 - MFCC feature (12-MFCC+E+delta+double delta)
 - GMM uses 512 Gaussian Components
- **GMM-MFCC systems (s2)**
 - MFCC feature (12-MFCC+E+delta+double delta)
 - UBM with 512 Gaussian Components
 - I-vector dimension of 150, Fisher Discriminant Analysis
- **GMM-Prosodic system (s3)**
 - Prosody features: Pitch+ energy + duration features
 - GMM-UBM with 256 components.

System	Unweighted recall
S1 : GMM w/ MFCC	69.72%
S2 : Ivector w/ MFCC	69.81%
S3 : GMM-UBM w/ long-term feature	66.61%
S1+S2+S3	70.54%

I-vector modelling in Speaker recognition

Baseline results: Comparison of JFA and i-vector systems on the common subset of the **2008 NIST SRE** database. **WCCN**: Within-class Covariance Normalisation, **LDA**: Linear discriminant analysis, **SDNAP**: scatter-difference Nuisance Attribute Projection, **GPLDA**: Gaussian Probabilistic LDA

Utterance Length (training-testing)	JFA System		LDA + WCCN		SDNAP + WCCN		GPLDA System	
	EER	DCF	EER	DCF	EER	DCF	EER	DCF
2 sec - 2 sec	35.25%	0.0988	35.35%	0.0986	35.67%	0.0999	36.16%	0.0999
4 sec - 4sec	30.48%	0.0934	31.05%	0.0966	30.23%	0.0968	31.30%	0.0991
8 sec - 8 sec	23.39%	0.0803	23.95%	0.0800	23.56%	0.0801	23.56%	0.0837
10 sec - 10sec	21.17%	0.0738	21.56%	0.0741	20.84%	0.0737	20.34%	0.0762
20 sec - 20sec	12.79%	0.0533	13.41%	0.0530	12.84%	0.0528	11.87%	0.0532
50 sec - 50 sec	6.51%	0.0266	6.44%	0.0310	6.42%	0.0299	5.77%	0.0272
full (2.5min) - full	3.37%	0.0149	3.54%	0.0179	3.62%	0.0166	3.13%	0.0168

EER: Equal error rate, DCF: Decision Cost Function is equivalent to pairwise loss function (C_{avg})

Automatic speech recognition

- **Building of ASR system required various knowledge sources:**
 - **Acoustics** - knowledge about variability in speech
 - **Phonetics** - knowledge about characteristics of speech sounds
 - **Phonology** - knowledge about variability of speech sounds
 - **Prosodics** - knowledge about stress and the intonation patterns
 - **Lexical** - knowledge about patterns of language
 - **Syntax** - knowledge about the grammatical structure of language
 - **Semantics** - knowledge about the meaning of the words
 - **Pragmatics** - knowledge about the context of conversation

Automatic Speech recognition (ASR): is a transduction of spoken acoustic sequence to text sequence.



Automatic speech recognition

- Speech Recognition Problem: $P(W | Y)$
 - Y represents sequence of observation symbols (acoustic features MFCC), W represents the sequence of words.

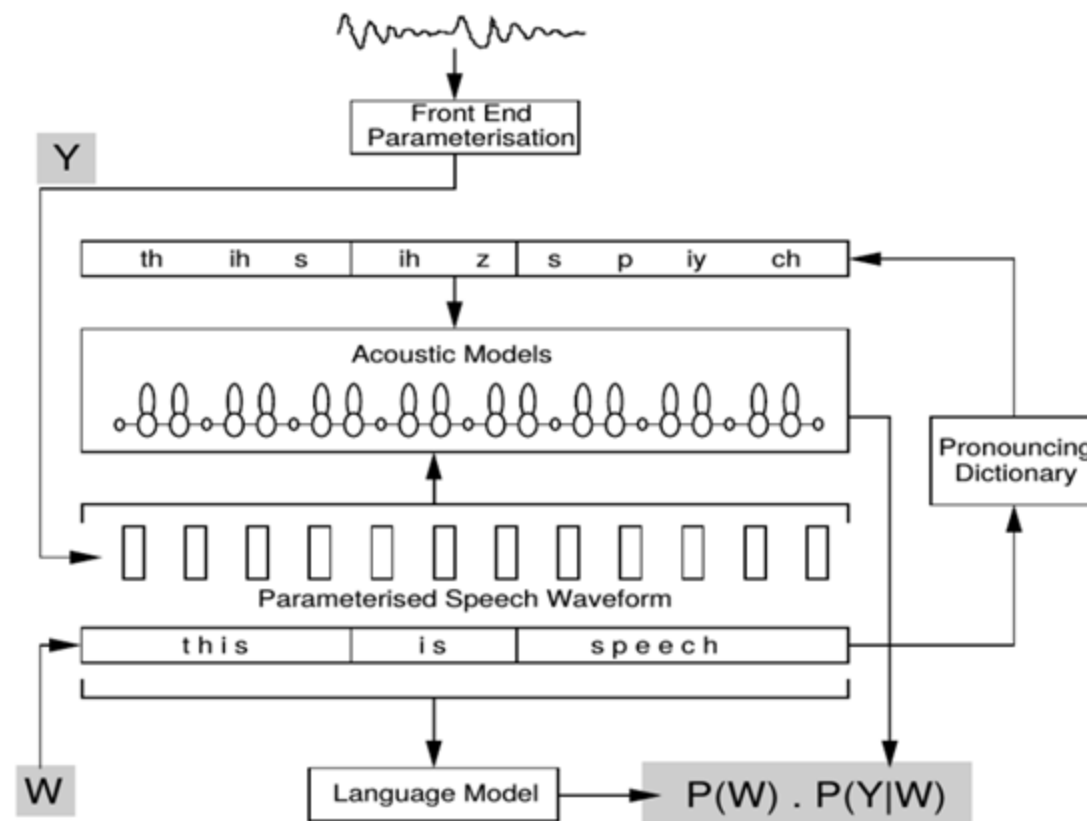
- Objective: maximize $P(W | Y)$ during training

Bayesian formulation for speech recognition:

$$P(W|Y) = P(Y|W)P(W)/P(Y)$$

$$\hat{W} = \arg \max_W P(W|Y) = \arg \max_W \frac{P(W)P(Y|W)}{P(Y)}$$

- $P(Y | W)$: likelihood function, $P(W)$: a priori probability distribution
 - Performance of Speech Recognition Systems: **word error rate (WER)** = $(S+I+D)/N$ Here S, I, D, C are the number of **substitutions**, **insertions**, **deletions** and **correct** words and N is (S + D + C)



Automatic speech recognition

- Widely used acoustic models:
 - GMM-HMM based acoustic models
 - DNN-HMM based acoustic models
 - RNN-CTC based acoustic models
 - Encoder-Decoder acoustic models

Automatic speech recognition

- **Hidden Markov models:**

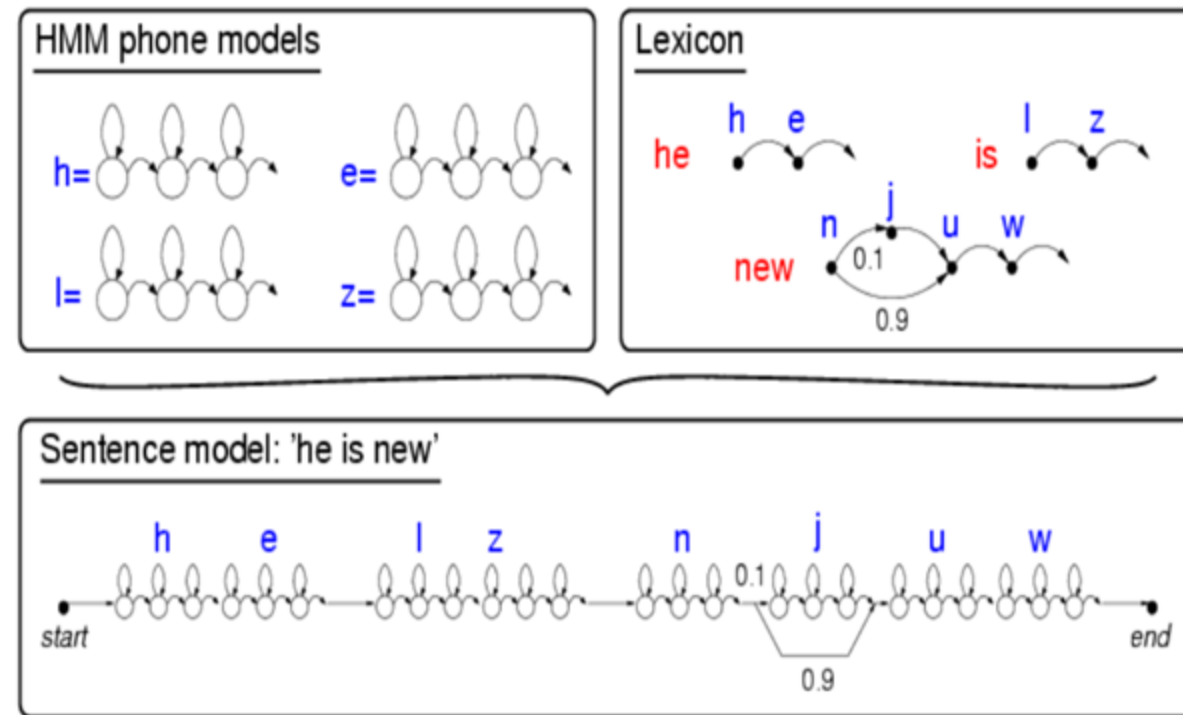
- A Markov chain is useful when we need to compute a **probability for a sequence** of observable events.

- **Three basic problems in HMM modelling:**

- **The Evaluation Problem:** Given an HMM and a sequence of observations $O = O_1, O_2, O_3, \dots O_T$ what is the probability that the observations are generated by the model, $P(O|\lambda)$?
- **The Decoding Problem:** Given a model λ and a sequence of observations $O = O_1, O_2, O_3, \dots O_T$ what is the most likely state sequence in the model that produced the observations? i.e $Q^* = q_1, q_2, q_3, \dots q_T$; $Q^* = \operatorname{argmax}_Q P(Q|O; \lambda)$; Here, $q_1, q_2, \dots$ are referred to states.
- **The Learning Problem** Given a model λ and a sequence of observations $O = O_1, O_2, O_3, \dots O_T$ how should we adjust the model parameters $\lambda = (\mathbf{A}, \mathbf{B}, \boldsymbol{\pi})$ in order to maximize $P(O|\lambda)$

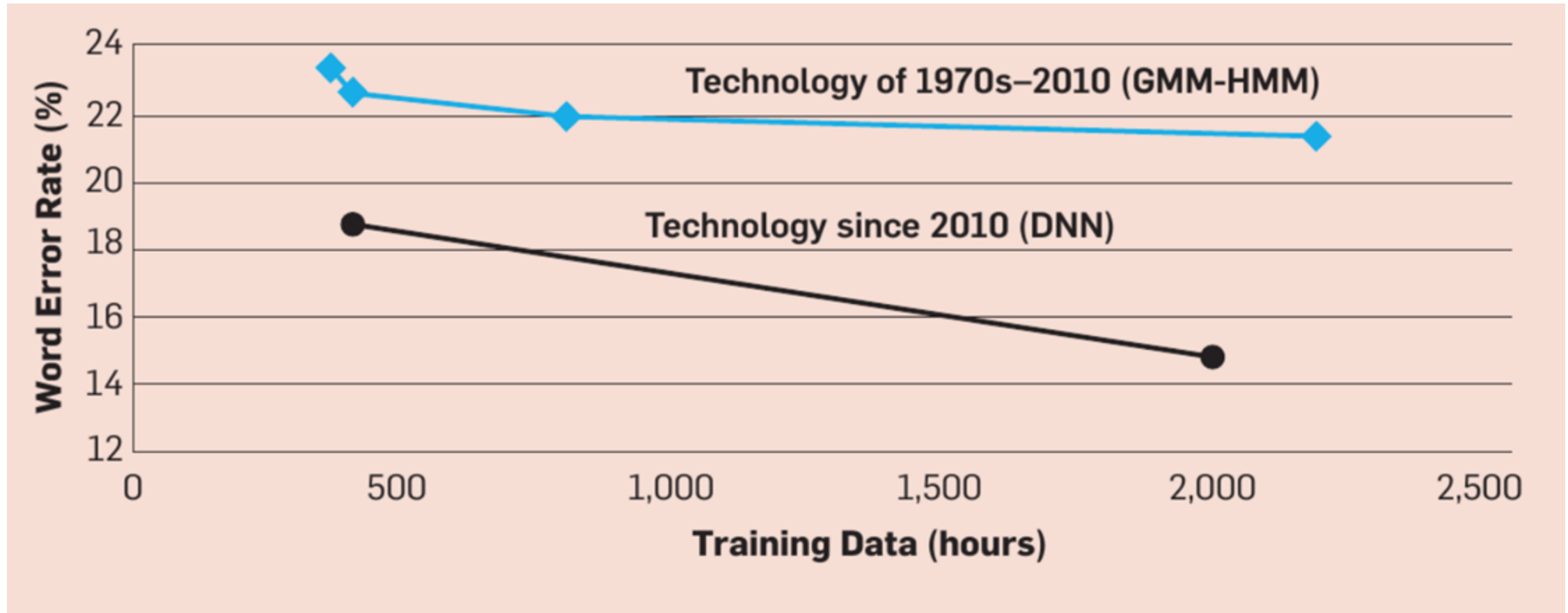
Automatic speech recognition

- **HMM-GMM based acoustic models**
- Spoken word can be decomposed into sequence of K_w basic phones (units) and the sequence called **pronunciation sequence**.
- HMMs model the temporal variability of speech and GMMs model how well each frame or a short window of frames fits a state of HMM
- In practice each phone is represented by a HMM with **left to right topology** and **three hidden states**.



Block diagram describing HMM-GMM based acoustic modeling.

Automatic speech recognition



Improvements in the performances of speech recognition systems with relevance to acoustic models (GMM-HMM, and DNN-HMM) as function training data

Automatic speech recognition

Performances of speech recognition systems developed using the [WSJ corpus](#).

Acoustic modeling approaches	LM	%WER (14 Hrs)	%WER (81 Hrs)
CD-HMM-GMM ¹⁵	Trigram	-	12.42 (4)
	None	-	-
CD-HMM-DNN ¹⁵	Dictionary	56.1	51.1
	Monogram	23.4	19.9
	Bigram	11.6	9.4
	Trigram	9.4	7.8 (3.5)

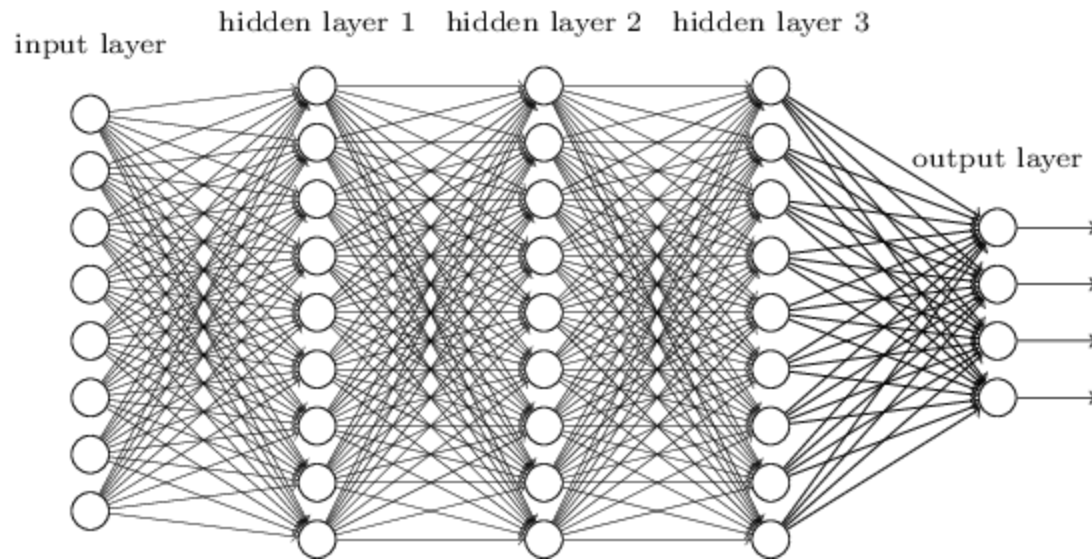
Thank you

Part-II

Advanced AI & ML in Speech Systems

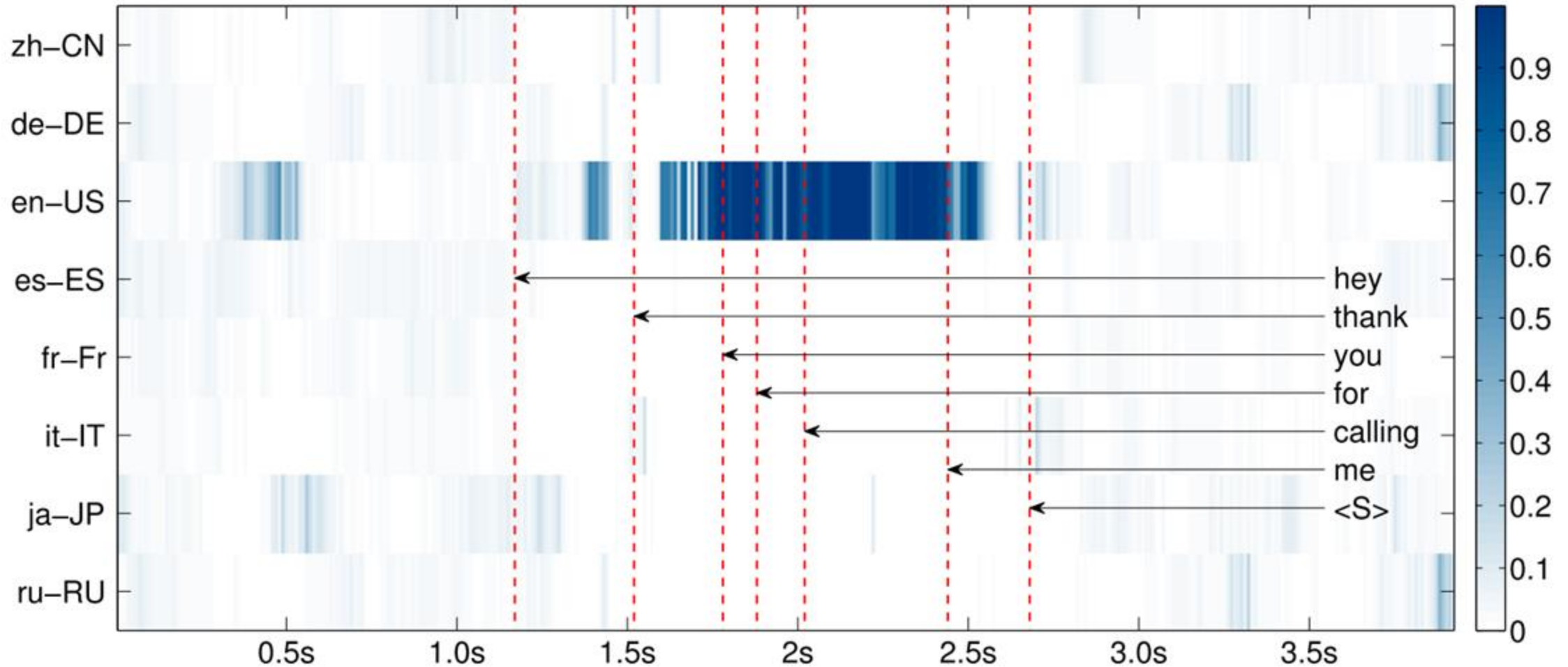


Deep Neural Network



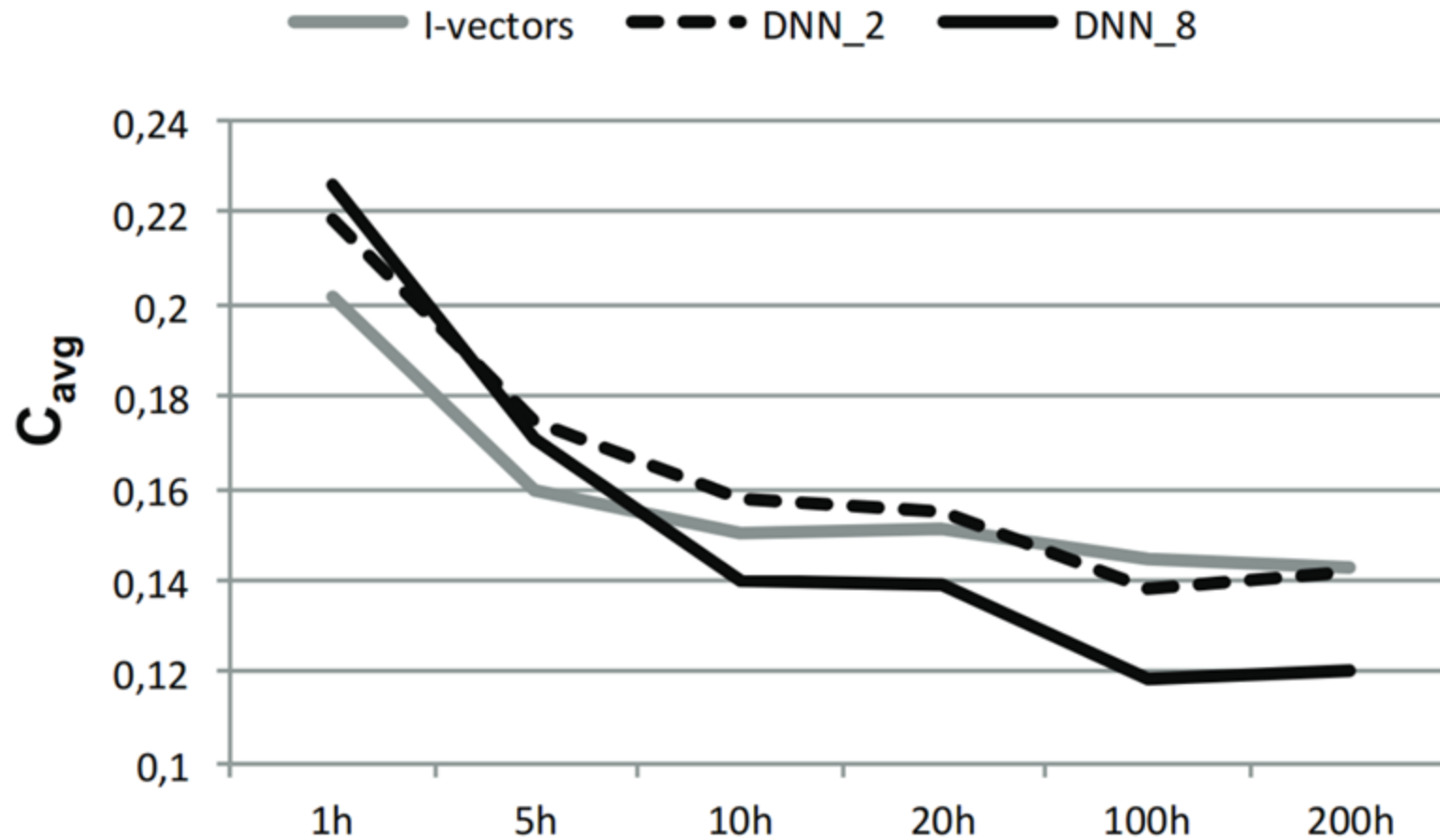
- Decision is taken at frame level.
- The frame level decisions are averaged to get utterance level decision.

Deep Neural Network



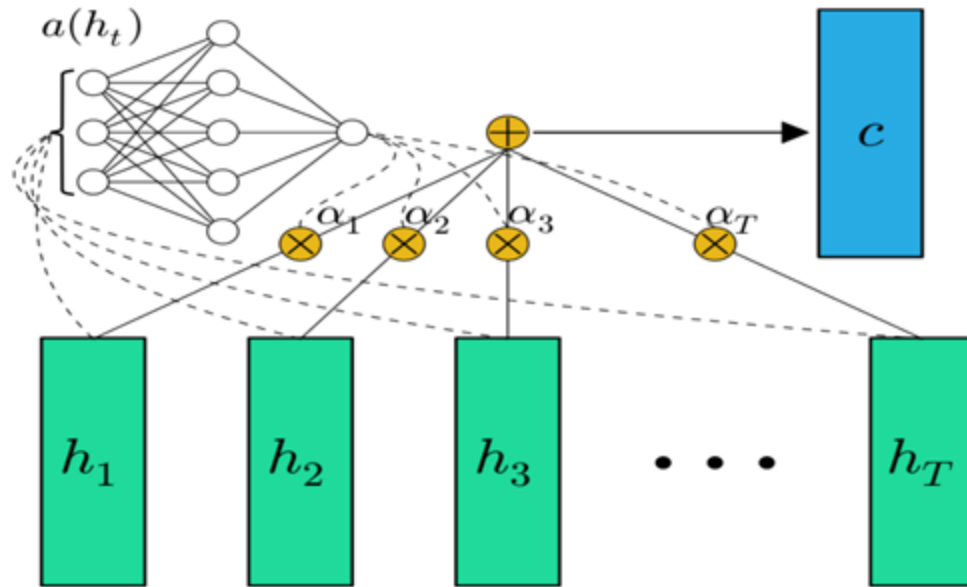
Frame level probabilities of a DNN-based LID system (8 languages selected) evaluated over an English-USA (4s) test utterance.

Deep Neural Network



i-vector vs DNN performance on LRE09 database. C_{avg} =average cost.

DNN with attention

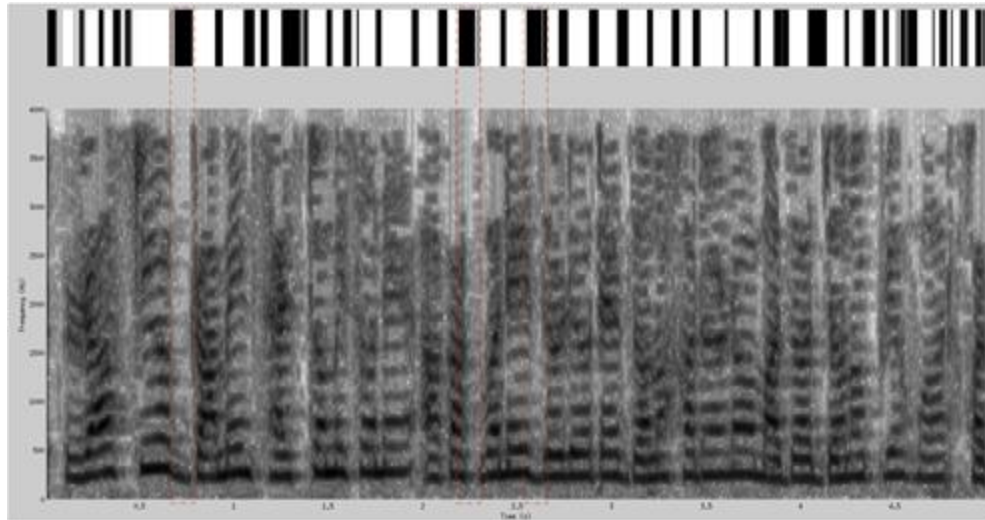


DNN with attention architecture

Context vector (c) is weighted average of hidden representations

- All frames may not contribute for decision / classification
- Decision is taken at utterance level.
- Weighted average of hidden representations

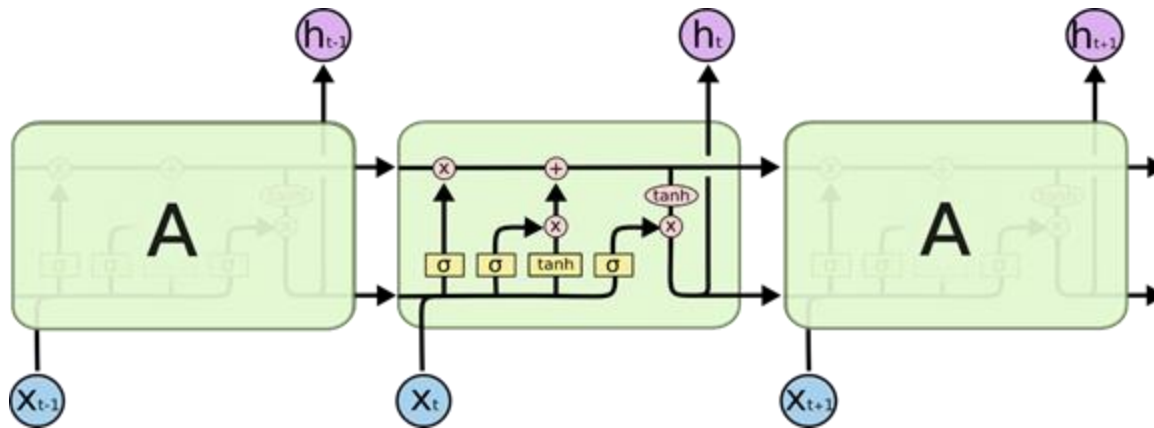
DNN with attention



An example of spectrogram with attention

- Attention weights are low for silence frames

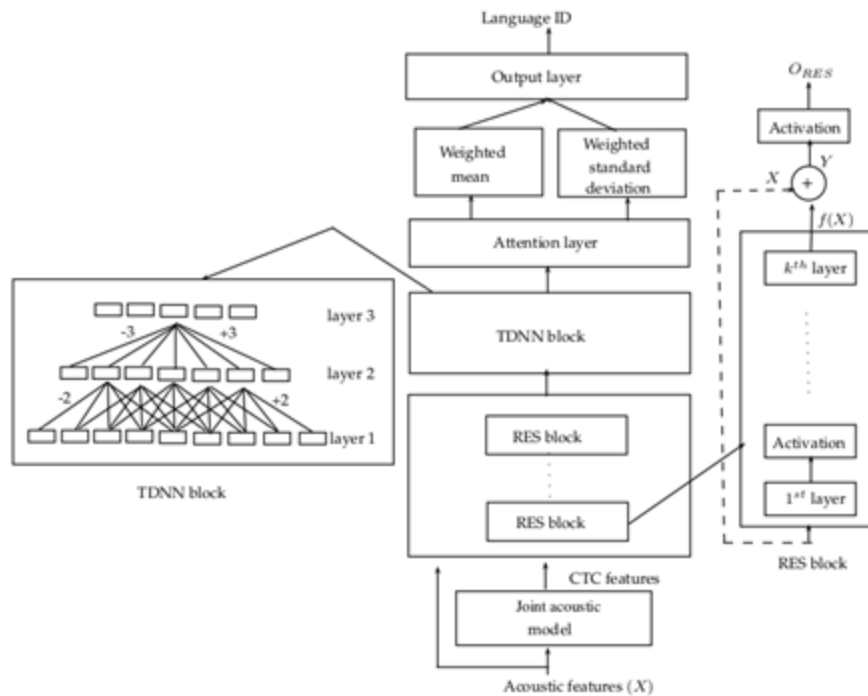
Sequential Networks



LSTM cell

- Long temporal dependencies
- Sequence to sequence mapping

Attention based residual time delay neural network



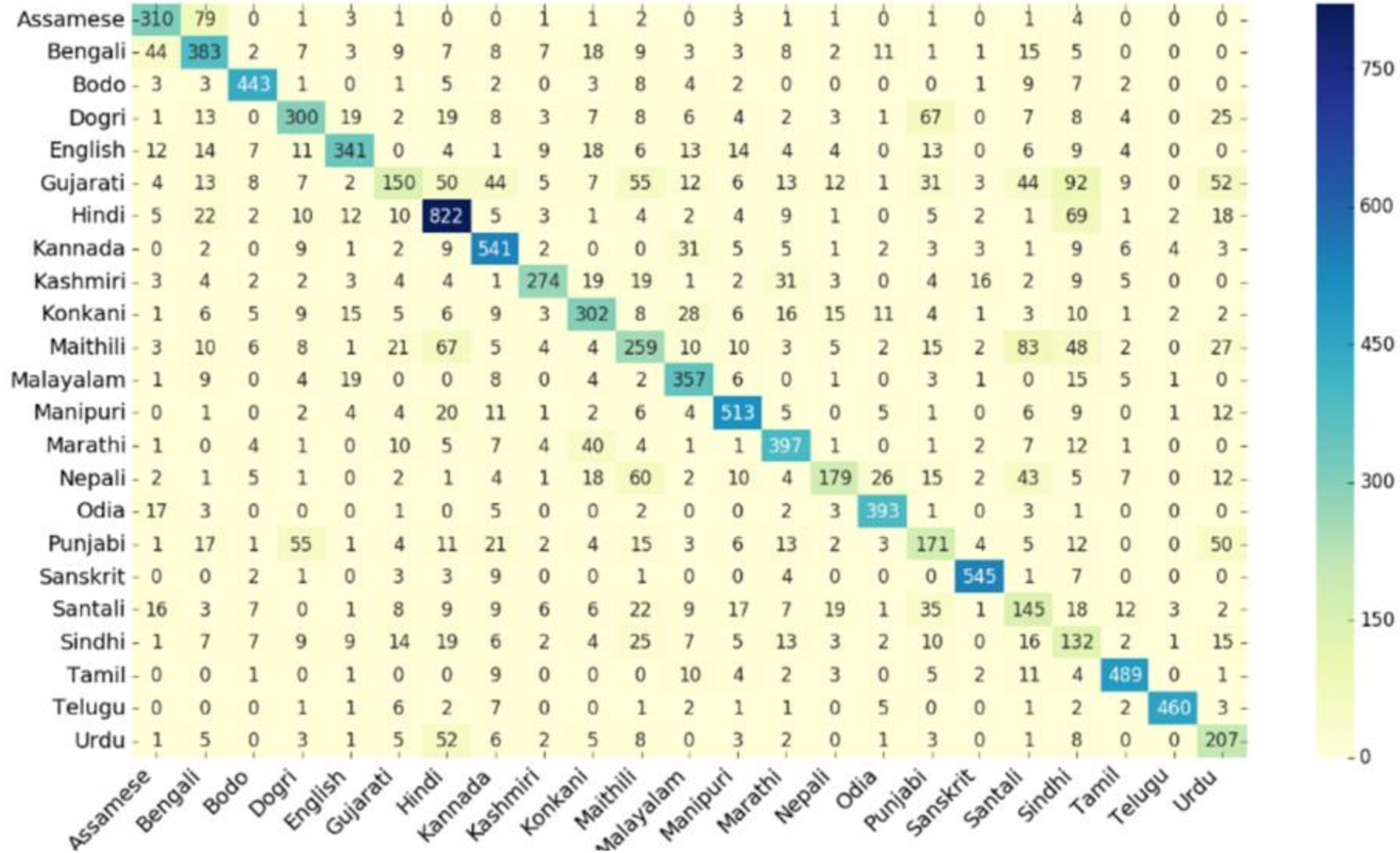
- Residual blocks allows skip connections which provide smooth flow of gradients
- In TDNN each layer captures temporal dependencies at different context
- Attention aggregates the whole input sequence information

Performance of Indian LID system using DNNs

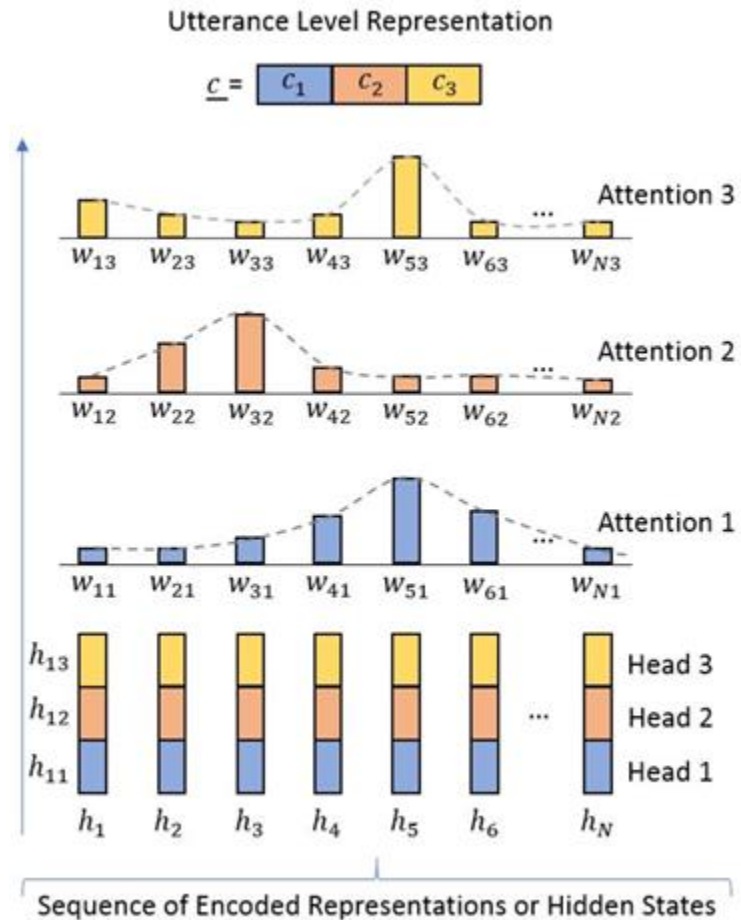
Network	Features			
	MFCC	SDC	i-vector	Phonetic
DNN	22.42	17.95	14.72	13.34
DNN-RES	21.87	17.12	14.25	12.56
LSTM	20.05	16.59	14.08	12.22
Attention based RES-TDNN	15.45	13.81	13.68	9.46

Results on IIITH-ILSC database using different neural networks

Performance of Indian LID system using DNNs

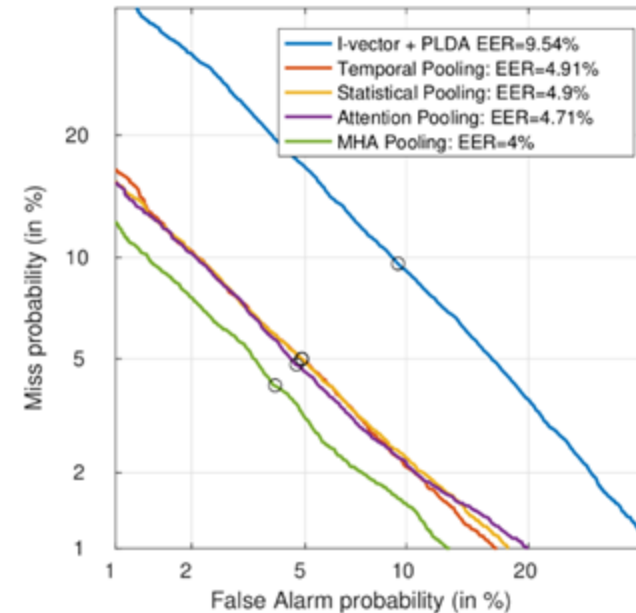


Performance of Speaker verification using Attention Networks



An example of the Self Multi-Head Attention Pooling with 3 heads

Approach	DCF	EER
I-vector + PLDA	0.0078	9.54
CNN + Temporal Pooling	0.0047	4.91
CNN + Statistical Pooling	0.0046	4.9
CNN + Att. Pooling	0.005	4.71
CNN + MHA Pooling	0.0045	4.0



Evaluation results of the text-independent verification on VocElb database.

Performance of Speaker identification using Attention Networks

The results for speaker identification on VoxCeleb

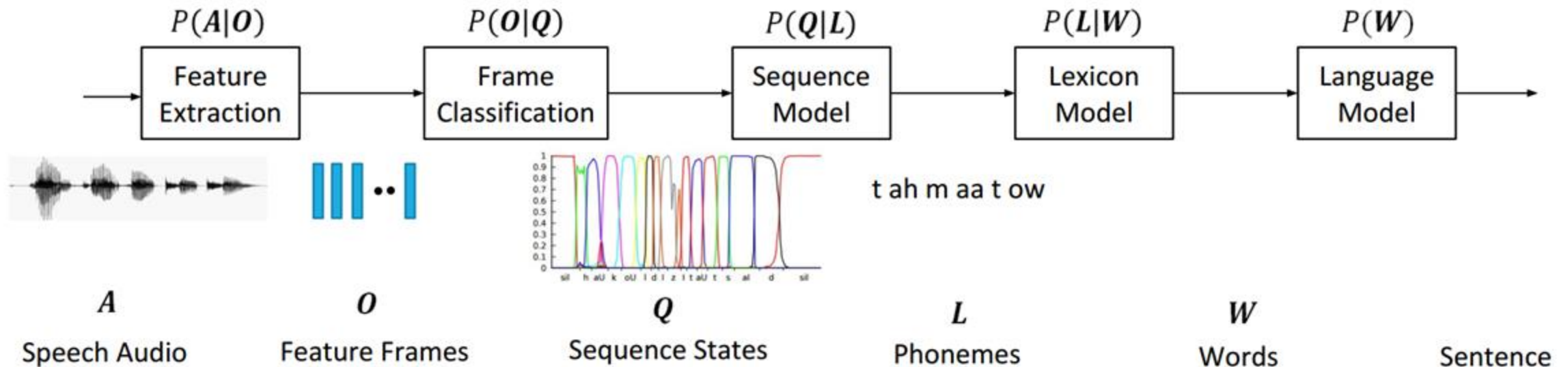
Accuracy	Top-1 (%)	Top-5(%)
I-Vectors + SVM [5]	49.0	56.6
I-Vectors + PLDA + SVM [5]	60.8	75.6
I-Vectors + LogReg [6]	65.8	81.4
VGG-like CNN+ TAP [5]	80.5	92.1
ResNet-34 + TAP [6]	88.5	94.9
ResNet-34 + SAP [6]	89.2	94.1
ResNet-34 + LDE [6]	89.9	95.7
VGG-like CNN+Self-Attention (ours)	88.2	93.8
ResNet-18+Self-Attention (ours)	90.8	96.5

The results of the three most frequently used acoustic features for speaker identification on VoxCeleb. Here, Spectr. corresponds to the spectrograms feature.

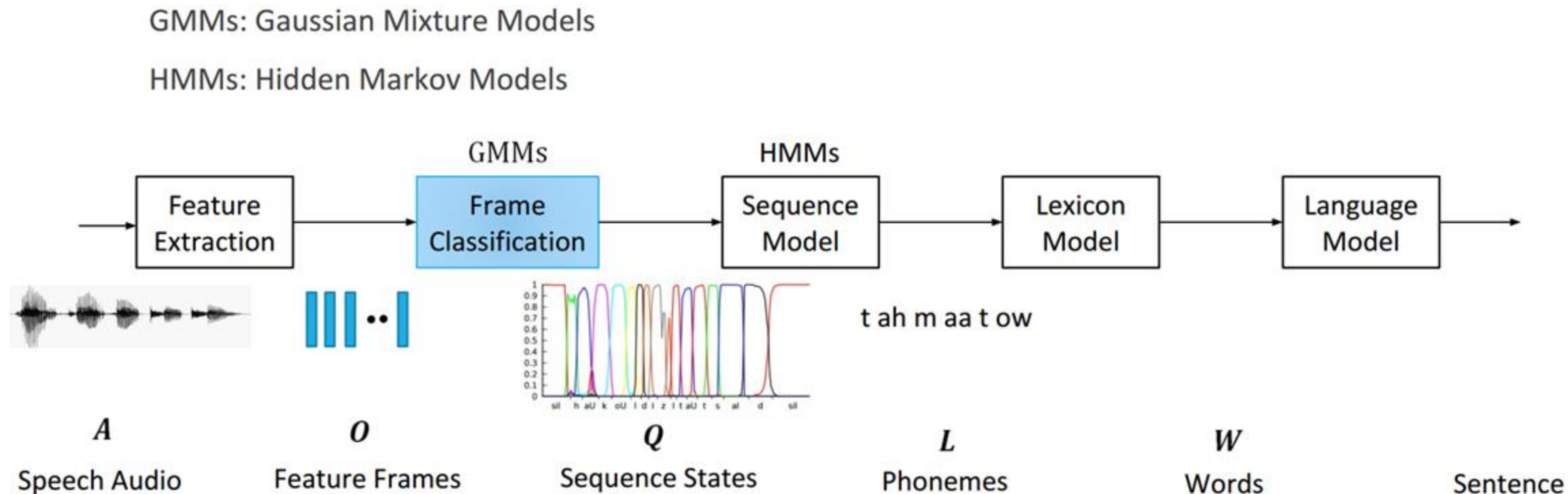
Model	Feature Type	Top-1 (%)	Top-5(%)
VGG-like CNN [5]	Spectr.	80.5	92.1
VGG-like CNN+Self-Attention (ours)	Spectr.	85.3	92.9
ResNet-18+Self-Attention (ours)	Spectr.	87.2	93.3
VGG-like CNN+TAP [5]	MFCCs	82.4	92.8
VGG-like CNN+Self-Attention (ours)	MFCCs	87.4	93.5
ResNet-18+Self-Attention (ours)	MFCCs	88.5	94.8
ResNet-34 [6]	FBank	89.9	95.7
VGG-like CNN+Self-Attention (ours)	FBank	88.2	93.8
ResNet-18+Self-Attention (ours)	FBank	90.8	96.5

Automatic speech recognition (recap)

$$\hat{W} = \underset{W}{\operatorname{argmax}} P(W|\mathcal{O}) = \underset{W}{\operatorname{argmax}} P(A|\mathcal{O})P(\mathcal{O}|Q)P(Q|L)P(L|W)P(W)$$



Automatic speech recognition (recap)



Lexical model

Lexical modelling forms the bridge between the acoustic and language models

Prior knowledge of language

Mapping between words and the acoustic units (phoneme is most common)

Deterministic

Probabilistic

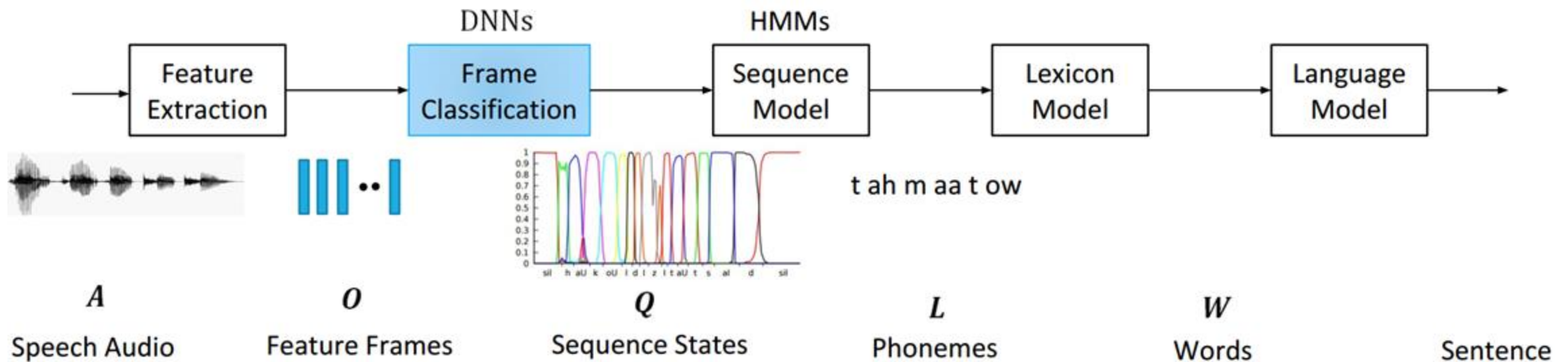
Word	Pronunciation
TOMATO	t ah m aa t ow
	t ah m ey t ow
COVERAGE	k ah v er ah jh
	k ah v r ah jh

Word	Pronunciation	Probability
TOMATO	t ah m aa t ow	0.45
	t ah m ey t ow	0.55
COVERAGE	k ah v er ah jh	0.65
	k ah v r ah jh	0.35

DNN-HMM in Speech Recognition

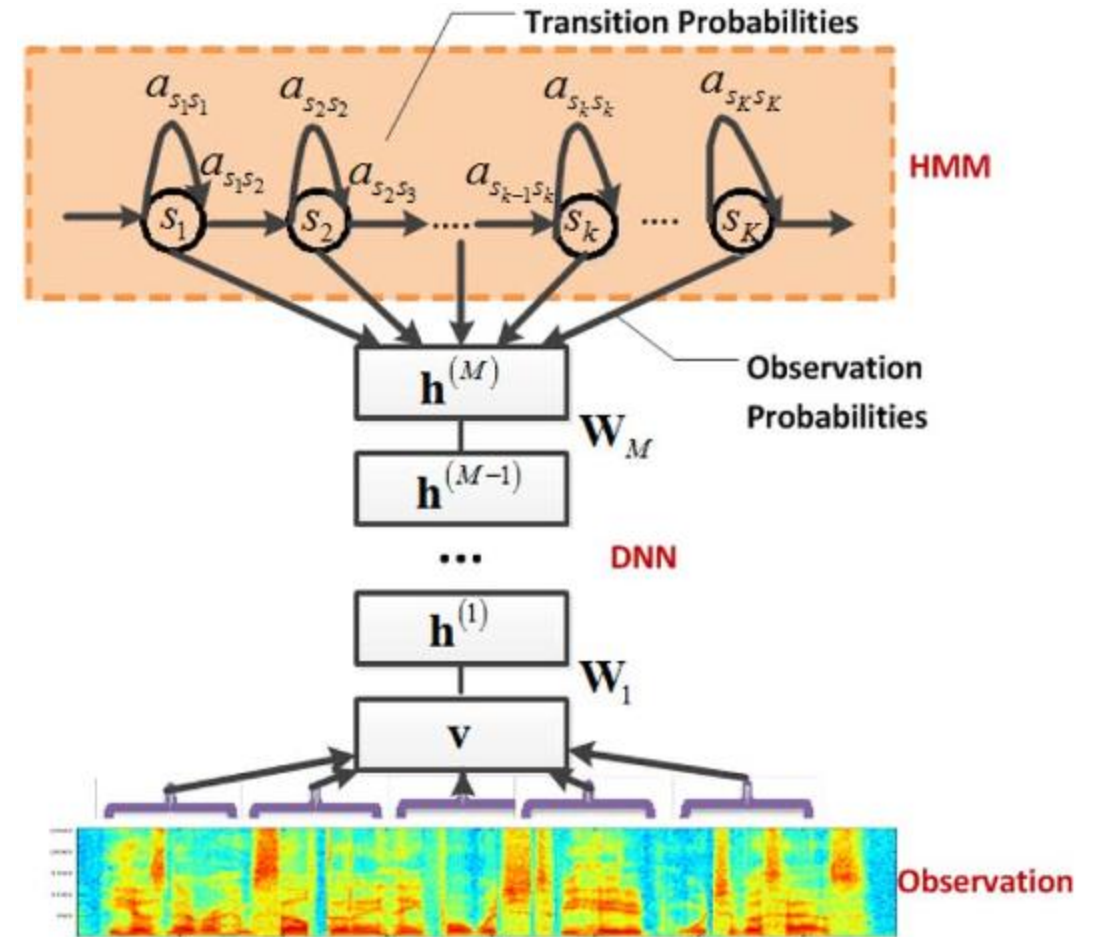
DNN: Deep Neural Networks

HMMs: Hidden Markov Models



DNN-HMM in Speech Recognition

- 1990s: Large vocabulary continuous dictation
- 2000s: Discriminative training (minimize word/phone error rate)
- 2010s: Deep learning significantly reduce error rate



George E. Dahl, et al. Context-Dependent Pre-Trained Deep Neural Networks for Large-Vocabulary Speech Recognition. IEEE Trans. Audio, Speech & Language Processing, 2012.

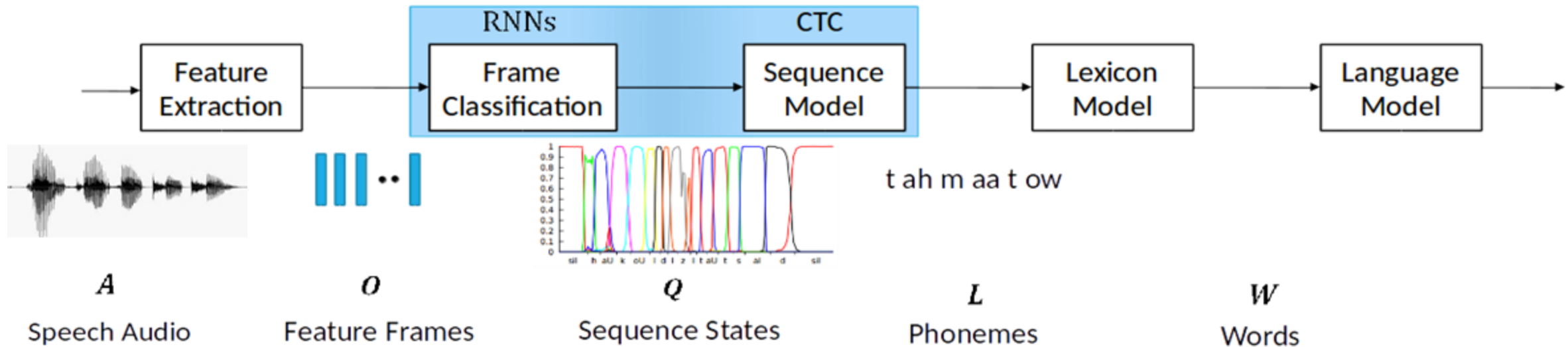
DNN-HMM vs. GMM-HMM

- **Deep models are more powerful**
 - GMM assumes data is generated from single component of mixture model
 - GMM with diagonal variance matrix ignores correlation between dimensions
- **Deep models take data more efficiently**
 - GMM consists with many components and each learns from a small fraction of data
- **Deep models can be further improved by recent advances in deep learning**

RNN-CTC in Speech Recognition

RNN: Recurrent Neural Networks

CTC: Connectionist Temporal Classification

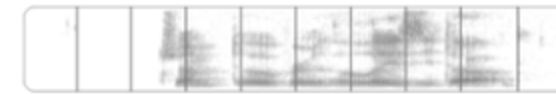


Connectionist Temporal Classification (CTC)

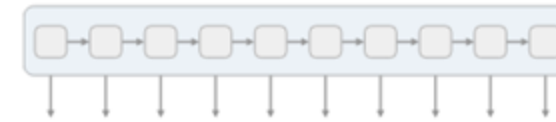
- No need of aligned data
- Reason: It can assign probability for any label, given an input
- It works by summing over the probability of all possible alignments between the input and the label
 - $\mathbf{X} = \{x_1, x_2, \dots x_T\}$ represents input sequences
 - $\mathbf{Y} = \{y_1, y_2, \dots y_T\}$ Transcripts
- Need an accurate mapping from X to Y
- Challenges using simpler supervised learning algorithms:
 - Both X and Y can vary in length.

Loss function

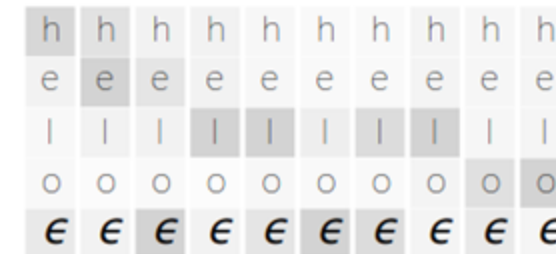
- The CTC alignments give us a natural way to go from **probabilities at each time-step** to the **probability of an output sequence**
- conditional probability
 - $P(Y|X) = \sum_{A \in A_{X,Y}} \prod_{t=1}^T p_t(a_t|X)$
- $p_t(a_t|X)$ computes the probability for a single alignment step-by-step.



We start with an input sequence, like a spectrogram of audio.



The input is fed into an RNN, for example.



The network gives $p_t(a_t|X)$, a distribution over the outputs $\{h, e, l, o, \epsilon\}$ for each input step.



With the per time-step output distribution, we compute the probability of different sequences



By marginalizing over alignments, we get a distribution over outputs.

LSTM-CTC MODELS: switch board corpus

Word error rate compared with HMM based models

	Context	Error Rate (%)
LSTM-HMM	Uni	8.9
	Bi	9.1
LSTM-CTC	Uni	9.4
	Bi	8.5

Deep learning in ASR on Chime and switch board

WER (SWB)	WER (full=SWB+CH)	Paper	Notes
5.5%	10.3%	English Conversational Telephone Speech Recognition by Humans and Machines	ResNet + BiLSTMs acoustic model, with 40d FMLLR + i-Vector inputs, trained on SWB+Fisher+CH, n-gram + model-M + LSTM + Strided (à trous) convs-based LM trained on Switchboard+Fisher+Gigaword+Broadcast
6.3%	11.9%	The Microsoft 2016 Conversational Speech Recognition System	VGG/Resnet/LACE/BiLSTM acoustic model trained on SWB+Fisher+CH, N-gram + RNNLM language model trained on Switchboard+Fisher+Gigaword+Broadcast
6.6%	12.2%	The IBM 2016 English Conversational Telephone Speech Recognition System	RNN + VGG + LSTM acoustic model trained on SWB+Fisher+CH, N-gram + "model M" + NNLM language model
8.5%	13%	Purely sequence-trained neural networks for ASR based on lattice-free MMI	HMM-BLSTM trained with MMI + data augmentation (speed) + iVectors + 3 regularizations + Fisher
9.2%	13.3%	Purely sequence-trained neural networks for ASR based on lattice-free MMI	HMM-TDNN trained with MMI + data augmentation (speed) + iVectors + 3 regularizations + Fisher (10% / 15.1% respectively trained on SWBD only)
12.6%	16%	Deep Speech: Scaling up end-to-end speech recognition	CNN + Bi-RNN + CTC (speech to letters), 25.9% WER if trained only on SWB

HMM based system still performs better than End-to-End system on large scale dataset

Part-III

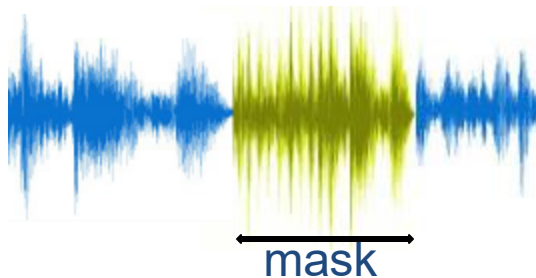
Advanced E2E -ASR models

Self Supervised Learning(SSL):

What is SSL:

- Uses one part of the input as labels to learn representations.
- Does not require manually annotated labelled data which is time consuming and expensive.
- Allows training on **unlabelled** data which is available in large quantities from the internet.

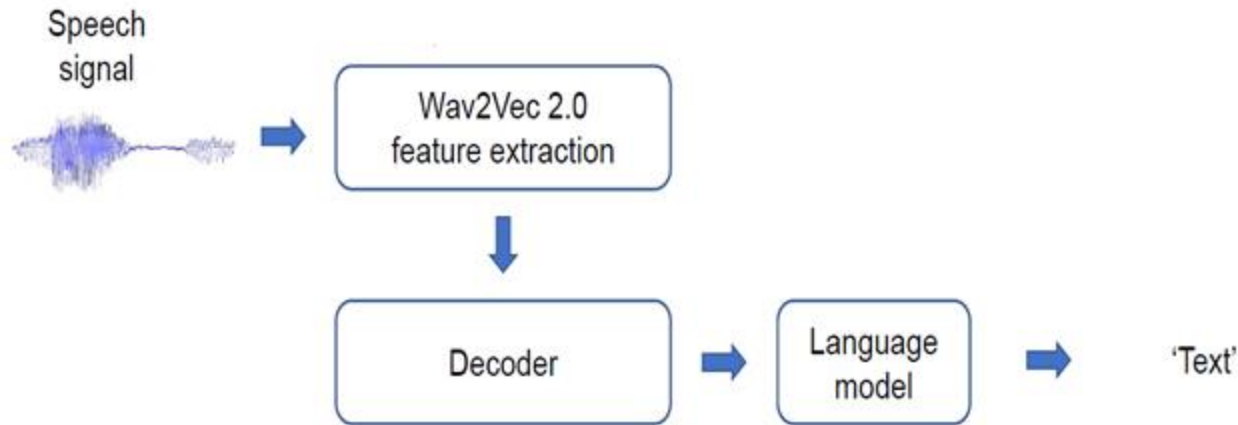
How SSL works on speech signals:



- Masked part of the signal act as labels
- In training, the model will learn to predict the masked parts. In this way meaningful representations will be learnt.

Wav2Vec 2.0:

- Acts as a **feature extractor**
- Uses **SSL** method to learn speech representations from raw audio input.
- Training objective is based on **contrastive loss**, which maximizes the similarity between masked regions and the predicted regions.
- Model building in Wav2Vec follows 2 steps:
 - Pretraining - to learn speech representations by SSL learning
 - Fine-tuning - Supervised training based on different downstream tasks such as ASR, emotion recognition, Language identification, etc.,

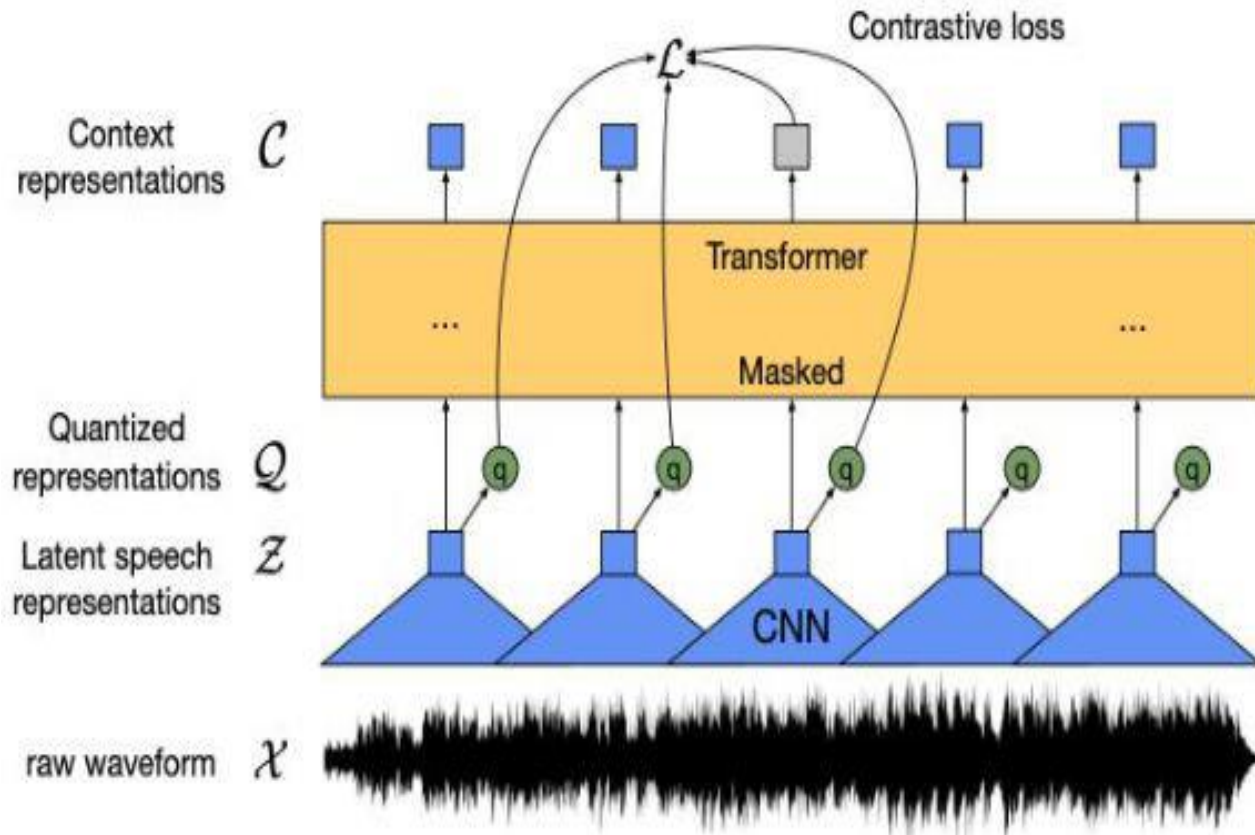


Advantages:

- Provides better results for small amount of data
- Shows significant improvement on **low resource** asr
- Learns **multilingual** representations
- Impressive performances on wide range of downstream tasks.

Wave2Vec2.0 based ASR

Wav2Vec 2.0 architecture:



Wave2Vec2.0 architecture

- Architecture mainly consists of 4 main parts:
 - Feature Encoder: Converting raw waveform into sequence of feature vectors $Z_0, Z_1, Z_2, \dots, Z_T$ (Latent speech representations).
 - Quantization Module: Maps continuous-valued embeddings to discrete symbols using vector quantization.
 - Context Network: Improves the quality of the learned contextual embeddings generated by the feature encoder by capturing both local and global contextual information in the audio.
- Uses Gumbel softmax distribution, CTC loss
- Wav2vec2- XLSR (cross lingual speech representations) for multilingual speech recognition, trained on 128 languages.

Wav2Vec 2.0 results:

Model	Unlabeled data	LM	dev		test	
			clean	other	clean	other
10 min labeled						
BASE	LS-960	None	46.1	51.5	46.9	50.9
		4-gram	8.9	15.7	9.1	15.6
		Transf.	6.6	13.2	6.9	12.9
LARGE	LS-960	None	43.0	46.3	43.5	45.3
		4-gram	8.6	12.9	8.9	13.1
		Transf.	6.6	10.6	6.8	10.8
LARGE	LV-60k	None	38.3	41.0	40.2	38.7
		4-gram	6.3	9.8	6.6	10.3
		Transf.	4.6	7.9	4.8	8.2
1h labeled						
BASE	LS-960	None	24.1	29.6	24.5	29.7
		4-gram	5.0	10.8	5.5	11.3
		Transf.	3.8	9.0	4.0	9.3
LARGE	LS-960	None	21.6	25.3	22.1	25.3
		4-gram	4.8	8.5	5.1	9.4
		Transf.	3.8	7.1	3.9	7.6
LARGE	LV-60k	None	17.3	20.6	17.2	20.3
		4-gram	3.6	6.5	3.8	7.1
		Transf.	2.9	5.4	2.9	5.8
10h labeled						
BASE	LS-960	None	10.9	17.4	11.1	17.6
		4-gram	3.8	9.1	4.3	9.5
		Transf.	2.9	7.4	3.2	7.8
LARGE	LS-960	None	8.1	12.0	8.0	12.1
		4-gram	3.4	6.9	3.8	7.3
		Transf.	2.9	5.7	3.2	6.1
LARGE	LV-60k	None	6.3	9.8	6.3	10.0
		4-gram	2.6	5.5	3.0	5.8
		Transf.	2.4	4.8	2.6	4.9
100h labeled						
BASE	LS-960	None	6.1	13.5	6.1	13.3
		4-gram	2.7	7.9	3.4	8.0
		Transf.	2.2	6.3	2.6	6.3
LARGE	LS-960	None	4.6	9.3	4.7	9.0
		4-gram	2.3	5.7	2.8	6.0
		Transf.	2.1	4.8	2.3	5.0
LARGE	LV-60k	None	3.3	6.5	3.1	6.3
		4-gram	1.8	4.5	2.3	4.6
		Transf.	1.9	4.0	2.0	4.0

Figure: results on different sets of Librispeech

Whisper:

- Based on **weakly supervised learning**, learning without detailed annotation.
- Encoder- decoder architecture
- Trained on **680k hours** of **multilingual** audio data and their transcriptions, of which 117k hours are not in English and are belongs 96 languages

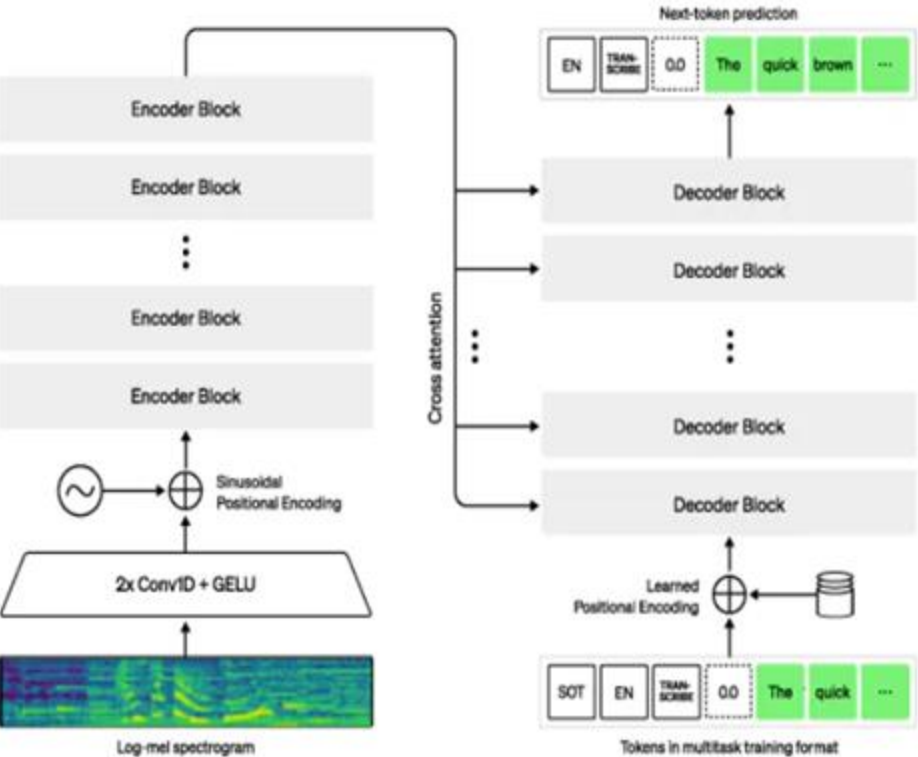


Figure 1:Architecture of Whisper

Dataset	wav2vec 2.0 Large (no LM)	Whisper Large V2
LibriSpeech Clean	2.7	2.7
Artie	24.5	6.2
Common Voice	29.9	9.0
Fleurs En	14.6	4.4
Tedlium	10.5	4.0
CHiME6	65.8	25.5
VoxPopuli En	17.9	7.3
CORAAL	35.6	16.2
AMI IHM	37.0	16.9
Switchboard	28.3	13.8
CallHome	34.8	17.6
WSJ	7.7	3.9
AMI SDM1	67.6	36.4
LibriSpeech Other	6.2	5.2
Average	29.3	12.8

Figure 2:Comparison of WER

Thanks!!

Questions?